535/1
PHYSICS THEORY
Paper 1
Jul. /Aug. 2026
2 1/2 hours

JINJA JOINT EXAMINATIONS BOARD

Uganda Certificate of Lower Secondary Education

MOCK EXAMINATIONS – AUGUST 2026

PHYSICS

Paper 1

2 hours 15 minutes

INSTRUCTIONS TO CANDIDATES:

This paper consists of **two** *sections*; **A** *and* **B**. *It has* **seven** *examination items.*

Section **A** *has* **three** *compulsory items.*

Section **B** *has* **two** *parts*; **I** *and* **II**. *Answer* **one** *item from each part.*

Answer **five** *items in all.*

Any additional question(s) answered will not be scored.

All answers **must** *be written in the booklets provided.*

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SECTION A

Answer **all** *the items in this section.*

Item one

Most people with eye defects prefer to regularly pay a visit to the Opticians for routine check-ups.

Jennie with her 15-year-old grandson, went to a shop that sells spectacles and ordered for a spectacle with lenses whose recommended power should not be less than **+8.29 D** for her to see clearly. The spectacles were provided but Jennie wants to know if they can work for her. Her grandson was curious about the type of the lenses his grandmother bought, the eye defect she was suffering from and how the defect is corrected by the lenses purchased.

In the nearby test laboratory, monkeys were being quarantined in a sealed glass container one at a time and making a lot of noise. This was done by the doctors to determine if the monkeys were healthy such that they test the vaccines. The sound of a healthy monkey should not be of more than **1710 Hz**. The doctor would turn on a certain pipe. Jennie and her grandson were left puzzled and asked the doctor to explain to them why the monkey's noise reduced gradually and suddenly disappeared when the pipe was turned on, even when the monkey's mouth was seen opening.

Jennie's grandson went for a short-call in the wash rooms of the laboratory and he could not understand how the curved mirror he was standing close to in the washroom produced an upright image smaller than his actual size. This type of mirrors has a number of applications in daily life situations. You have been contacted by the doctor and Jennie.

Hint:

When one of the lenses in the spectacles was placed 20 cm from an object, a clear sharp image was formed on a white screen 30 cm away from the lens.

Wavelength of the sound produced by the monkeys was 0.2 m.

Speed of sound in air is 340 m s $^{-1}$.

Task:

(a) Help;

(i) Jennie and her grandson to address their concerns on the spectacles and explain to them why the monkey's noise disappeared.
(*include illustrations where necessary*)

(ii) the doctor to find out if they can go ahead to test the vaccines.

(b) Explain any **two** daily applications of the mirror Jennie's grandson encountered in the washroom.

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Item two

On 1st April 2026, NASA sent a group of Astronauts (Artemis II crew) in a lunar flyby mission that lasted up to 10th April 2026. One of the pictures taken by the crew was used by a teacher in one of the secondary schools in Uganda for a lesson. The learners wanted to know why the earth had one part with light and the other side dark.

Support diagram

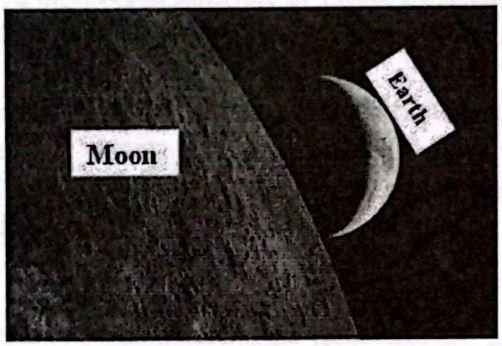


Photo taken by the crew on one side of the moon

A tourist from a winter-stricken nation flew to a certain ***island*** on 3rd April 2026 for a vacation. Upon his arrival, he noticed that the ***island*** was experiencing a very dry summer climate, despite being completely surrounded by an ocean. He could not understand the difference in seasons of his home country and the ***island***.
Furthermore, he also noticed a strange phenomenon, where in the morning hours, the ocean's water level was very low, but by the evening, the water level becomes very high.
On 9th April, the tourist prepared to visit Mpoma Satellite station in Uganda to clearly watch Artemis II flyback from the mission, but he had no idea how to locate the satellite station.
You were contacted for help, and you sent the tourist a navigation *app* (***Citymapper***) to install on his smart phone. The tourist watching the crew moving in the earth's atmosphere, was puzzled why the Capsule outer surface was emitting fire.

Task:

(a) Help the:

(i) students understand their observation.
(ii) tourist address his concerns when on the island.

(b) Explain to the tourist how the app can help him and explain why it is important for a developing nation like Uganda to invest in space exploration.

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Item three

As one of the countries embracing sustainable development, Uganda started implementing "*Green energy*" policy by use of electric buses, and has started plans to generate more electrical energy using nuclear processes in *Buyende* district near lake *Kyoga*.

The ministry of energy and mineral development extracted and installed a radioactive mineral element in the nuclear reactor to start the process of production of electricity. At the time of installation of the mineral element, its count rate was **1000 counts per minute.**
The "***Green energy***" experts advised the ministry that for the production process to be controllable, the production of electricity should start at the time when the count rate of the element is **10 counts per minute** above the background radiation. The ministry is searching for a person to help it to know, how long it should take for the production process to start.

The people living in the surrounding community are puzzled and wish to understand how the too much water that the ministry plans to pump into the reactor, together with the mineral element help in the production of electricity.
They also want to be educated on the economic effects that come with the constructed power plant, and the health hazards associated with the power plant that made them get displaced.

Meanwhile, during the installation process, one of the workers was struck by a falling object which broke the bone of his arm. **Two** of the community members rushed him to the nearby hospital where his arm was subjected to a device that uses highly penetrative radiations that takes picture of fractures. The residents were curious to know how the device works and how it is used to locate the fracture in the worker's hand.

You have been approached.

Hint:

***The half-life of the radioactive element is* 2 years.**
***The background radiation is* 45 counts per minute.**

Task:

(a) Help

(i) the ministry to establish the time electricity production will start.
(ii) and address the concerns of the surrounding community.

(b) Explain to the two community members the principle of operation of the device and the application they wish to understand.

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SECTION B

(Part I)

Respond to **one** *item only in this section.*

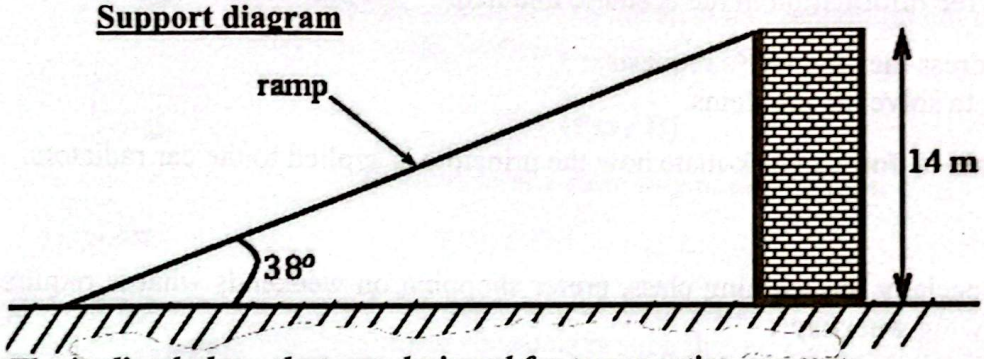
Item four

During the construction of a storeyed building near the lake shore, workers are made to lift the concrete of mass **250 kg** using an inclined plane and pulley system containing a pair of pulleys in each of the upper and lower blocks.

The Chief Engineer's Requests are;

(i) how the pulley system should be arranged and workers should use the system whose efficiency is not less than **71%**.

Support diagram



The inclined plane that was designed for transporting concrete

(ii) **Joby** monitoring the pulley system should always reach the pulley area at a velocity not less than **4.3 ms⁻¹.** The engineer wants to know if **Joby** will manage the task or whether an automated machine system should be installed to monitor the pulleys.

Joby upon reaching the pulley area, has to remove the lower ***hot*** pulley whose temperature is **110°C** and insert it quickly into **231*g*** of ice blocks placed in a metallic container such that the pulley cools after which he fixes the pulley back into its place if the temperature becomes not more than 37.8°C. He wishes to know whether to reduce or increase the ice blocks used.

Joby could not understand why;

(i) he became tired while climbing to remove the pulley but did not feel the same way when lowering the pulley into the ice blocks.
(ii) why he felt cool air blowing towards him when he was taking a walk along the lake shore.

Joby's *work-mate* was curious about the principle used in cooling the pulley and wants to know why water instead of cooking oil is used in car radiators.

You have been contacted for aid.

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Hint:

Job's workplace is **20 m** *away from the pulley area and takes him* **10 seconds** *to reach the pulley area.*
Specific latent heat of fusion of ice is $\mathbf{3.36 \times 10^5\ J\ kg^{-1}}$
The container in which the Ice blocks are placed weighs **3.0 kg**
Specific heat capacity of the container in which the ice is placed is $\mathbf{2100\ J\ kg^{-1}\ K^{-1}}$.
Force applied on each of the systems is **3750 N**.
Mass of each pulley in the lower block is **10 kg**.
Specific heat capacity of water is $\mathbf{4200\ J\ kg^{-1}\ K^{-1}}$.
Specific heat capacity of the material *of the pulley is* $\mathbf{490\ J\ kg^{-1}\ K^{-1}}$.
Acceleration due to gravity is $\mathbf{10\ m\ s^{-1}}$

Task:

(a) Read the information in the scenario and help

(i) to address the engineer's requests.
(ii) **Joby** to solve his problems.

(b) Explain to **Joby's** work-mate how the principle is applied to the car radiators.

Item five

Families, especially the working class, prefer shopping on weekends what is required for the following weekdays.

Nelkon and his daughter, Jalia, went to a storeyed shop to buy some goods. They observed the following phenomena:

(i) Nelkon, watching a movie on the shop TV screen, saw a gun moving backwards as the bullets were being fired out of the gun barrel.

(iii) after buying a balloon and inflating it, the balloon accidentally moved out of Jalia's hands and when the mouth of the balloon opened suddenly, it rushed upwards as the air moved out in the opposite direction.
These left both Nelkon and his daughter puzzled.

At the shop, there was giving prizes to a person who throws a wooden ball vertically upwards to reach a height of **10 m** above the ground level. Jalia, who is **140 cm** tall, was a participant. She threw the ball which took **1.2 seconds** to reach its maximum point. She wants to know if she won the prize and if not, what does she have to do next time?

Jalia's teacher requested her to boil a metallic steel ball in water to **100°C** and pour the water together with the ball in the flask and cover it. The following day Jalia will use the steel ball, which will be still at **100°C**, to explain to the class during the lesson, how the ball's specific heat capacity is obtained through a scientific investigation.
Jalia is concerned about the possibility of the steel ball remaining hot up to the lesson time, she also doesn't know how to carry out the scientific investigation requested by her teacher.

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You have been approached by Nelkon and his daughter, Jalia.

Hint:

Acceleration due to gravity is $\mathbf{10\ ms^{-2}}$

Task:

(a) Your work is to:

(i) Explain to Nelkon and his daughter, Jalia to understand the physics principles behind their observations.
(ii) Find out if Jalia won the prize and if not, what does she have to do to win it next time.

(b) Help Jalia:

(i) understand the possibility of the steel ball remaining hot up to lesson time.
(ii) to respond to the teacher's request in class.

(Part II)

Respond to **one** *item only in this section.*

Item six

A house wiring system requires careful consideration of the wires used and the appliances to be connected into the sockets such that no electrical damage occurs.

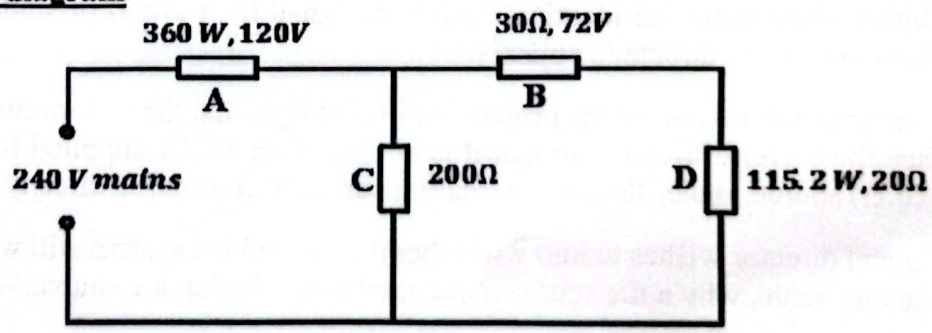
Johnson who has just rented a house has been told by his landlord that he has to pay **1655 US dollars** as a fee for electricity per **30 days.**

Johnson claimed that the amount quoted was a lot and requested the landlord to refund his rent such that he finds another house.

The landlord counter-claimed that Johnson had appliances of higher power ratings that would command the quoted amount but accepted to refund Johnson the rent if payment is found to be lower than what he requested Johnson to pay.

The landlord pulled out the connection plan showing the arrangement of Johnson's appliances **A, B, C** and **D** which are shown in the support diagram below.

Support diagram



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Circuit plan that showing the arrangement of Johnson's appliances

Johnson's son, who is in S.3, told the father that the house wiring cannot help the appliances to work properly and others would easily burn.
This angered the landlord who told Johnson that for the appliances to work properly, the former tenant in the same house had installed a device across each appliance to work at full voltage of the mains, should the current from the source be less than **3.5A**.

Johnson's son also complained that their house was not protected from lightning and requested the landlord to apply some knowledge of electricity to protect their rented house in case they don't leave the place.

You have been contacted by the landlord.

Hint

Cost of each unit of electricity is **9.6 US dollars**.
Every appliance works for **8 hours** *per day for a duration of* **30 days**.

Task:

Your work is to address the following engineering and financial queries:

(a) (i) Find out whether Johnson has to be refunded the rent, explain to him why the appliances are most likely to burn and how to minimize the effect.

(ii) Determine if Johnson needs to install the device suggested by the landlord and explain how it can help to make the appliances work properly.

(b) Explain how the landlord can apply the knowledge of electricity to protect his tenants and their properties as claimed by Johnson's son in the scenario.

Item seven

Metal scrap factories play an important role in recycling metal wastes that would otherwise cause pollution in the environment.

A technical director from one of the scrap factories is facing a critical issue involving delays in packing scrap metal into the crushing machine. He is searching for a person to design system capable of lifting a large volume of scrap from the off-load site to the crushing machine, and explain to the director how it works before he pays for it.
In response, a physics club from one of the secondary schools in Uganda has organized an exhibition showcasing an identical system designed by a group of students, which they believe meets the director's objectives.

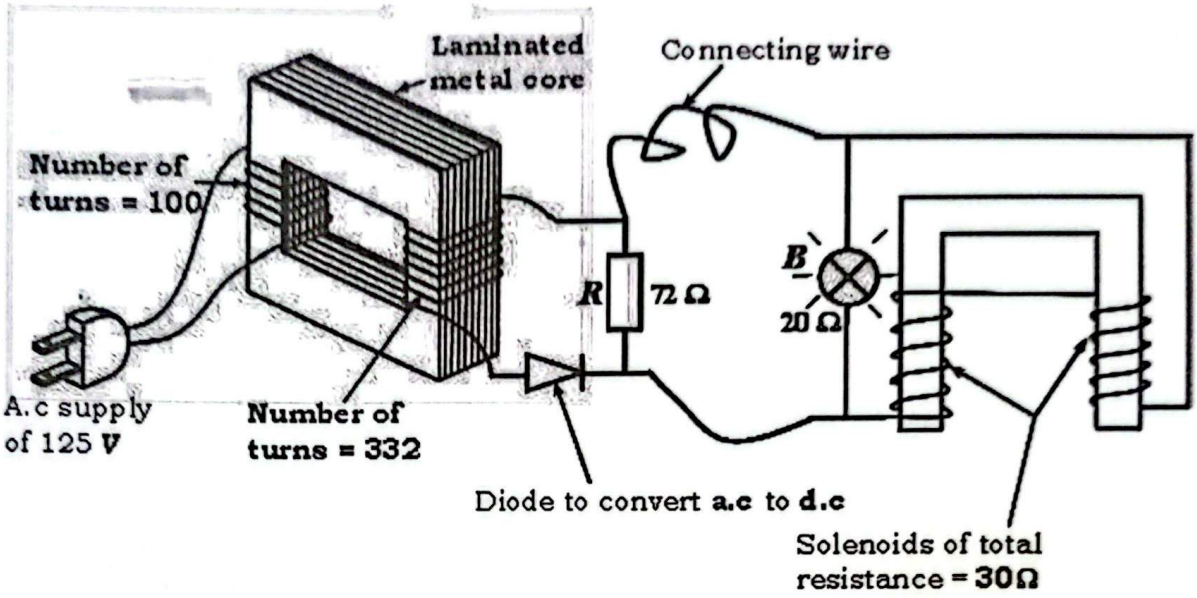
The technical specifications of the project report highlight that the system functions well only when the current flowing through it is not less than **14.2A** supplied from a direct current (**d.c.**) source, rather than an alternating current (**a.c.**) source directly.

The technical director wishes to know whether this assembled system will work reliably for his factory scale, why a **d.c** source is required instead of an **a.c** source.

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Support diagram



The system designed by the students. R is a protective resistor and B is a monitoring bulb.

The director also wants to know, in case the lifting mechanism is weak, what modifications can be made to strengthen the system to do the work in the shortest time possible.
Furthermore, workers in the factory have complained of overheating of the wires when the machine is running and require an investigation into the origin of this phenomenon.

You have been approached as part of the group carrying out the system project.

Task:

(a) Read the scenario and help to:

(i) Address the technical director's concerns.
(ii) Help the workers understand the origin of the phenomenon happening on the connection wires attached to the machine

(b) Explain any other **two** daily life applications of the scientific concepts used in the system setup described in the scenario.

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